

The \$50/mo AI SDR Stack

A step-by-step build guide. Follow it top to bottom and you'll have a working autonomous outreach system: it pulls leads, writes a personalized email for each, sends from a warmed inbox, reads the reply, and routes hot leads to you. Same architecture I build for clients instead of \$1,500-\$2,000/mo tools.

What you'll have at the end

- An n8n workflow running on a \$5 server that you can leave on 24/7.
- 3 warmed sending inboxes that land in the primary tab, not spam.
- A morning run that emails 60-90 fresh leads/day, each message personalized to one real fact.
- Auto-classified replies: interested leads ping you instantly, everything else self-sorts.

Before you start (accounts + budget)

What	Where	Cost	Time to set up
A VPS (small cloud server)	Hetzner / Hostinger / DigitalOcean	~\$5/mo	20 min
An LLM API key	Anthropic (Claude) or OpenAI	~\$20/mo usage	5 min
3 sending domains + mailboxes	Namecheap + Instantly	~\$15/mo (+~\$30 domains once)	30 min + 14-day warmup
A lead source	Apollo.io (free tier to start)	\$0-49/mo	15 min

Start the domains + warmup on day one. The 14-day warmup is the only thing you cannot rush, so kick it off first and build the rest while it runs.

Part 1 - Deploy n8n (~20 min)

n8n is the engine that runs the whole flow. Self-host it so runs are unlimited and free.

1. Create a VPS: pick the cheapest shared plan (1 vCPU / 2 GB RAM is plenty), Ubuntu 22.04.
2. SSH in and install Docker: `curl -fsSL https://get.docker.com | sh`
3. Run n8n with one command (persists your workflows to a local volume):

```
docker run -d --restart unless-stopped --name n8n -p 5678:5678 \
  -e N8N_HOST=<your-server-ip> -e N8N_SECURE_COOKIE=false \
  -e GENERIC_TIMEZONE=<your-timezone> \
  -v n8n_data:/home/node/.n8n docker.n8n.io/n8nio/n8n
```

Then open `http://<your-server-ip>:5678` in your browser and set your owner login.

For a real deployment put it behind a domain + HTTPS (Caddy or a reverse proxy). For testing, the IP is fine.

Part 2 - Domains, mailboxes + warmup (~30 min, then 14 days)

Never send cold email from your main domain. Buy 2-3 lookalike domains, put 1-2 mailboxes on each, warm them, then send.

1. Buy 2-3 domains close to your brand (e.g. `get-yourbrand.com`, `try-yourbrand.com`). ~\$10 each.
2. Set the 3 DNS records on each domain so you authenticate as a real sender:

```
SPF (TXT @): v=spf1 include:spf.your-email-provider.com ~all
DKIM (TXT): <paste the exact key your mailbox provider gives you>
DMARC (TXT _dmarc): v=DMARC1; p=none; rua=mailto:you@yourdomain.com
```

1. Create the mailboxes (Google Workspace, or a provider Instantly resells) - use a real first/last name, not "sales@".
2. Connect each mailbox to Instantly, turn ON warmup: ~20-40 warmup emails/day, ramping.
3. Wait 14 days. Do not send a single cold email before then. This is what keeps you out of spam.

Sending caps once warm: 20-30 real emails/day PER mailbox. 3 mailboxes = ~60-90/day. Going over is the #1 way people torch their domains.

Part 3 - Your lead source (~15 min)

1. In Apollo, build a saved search that IS your ICP: title, industry, headcount, geography.
2. Verify emails are included. Keep the list tight - 100 great-fit leads beat 1,000 sprayed.
3. Get an Apollo API key (Settings -> Integrations -> API), you'll call it from n8n. (Free tier works to start; you'll want a paid seat past ~1,000 leads/mo.)

Part 4 - Build the workflow, node by node

In n8n, create a new workflow and add these nodes in order. Each one's key settings are listed.

- 1. Schedule Trigger.** Add node -> Schedule Trigger. Set it to run once every morning (e.g. 8:00 AM). This kicks off the daily batch.
- 2. HTTP Request - fetch leads.** Add an HTTP Request node pointed at Apollo's search API (POST, your API key in the header). Return today's batch. Add a Limit node right after, cap at 60-90 so you never exceed your send ceiling.
- 3. Set - shape each lead.** Add a Set (Edit Fields) node. Map the fields you'll actually use: first_name, company, title, email, and one enrichment fact (see next node).
- 4. Enrichment fact.** The personalization hinges on ONE specific fact per lead. Pull it from Apollo's company fields (recent funding, headcount growth, tech used) or a second HTTP call to Clay. Store it as {{signal}}. One good line beats ten generic fields.
- 5. AI Agent (or Basic LLM Chain) - draft the email.** Add the AI Agent node, connect your Claude/OpenAI credential. Paste the Draft Prompt (Part 5). Feed it first_name, company, title, signal, and your one_line_offer. Output: the email body only.
- 6. IF - guardrail.** Add an IF node. Only continue if ALL are true: body is under ~120 words, the person's first name is present, and there are NO leftover placeholders like {{company}}. This one node stops you from mailing 200 people a broken merge.
- 7. Instantly / SMTP - send.** Add the send node (Instantly node or SMTP). Rotate across your 3 mailboxes so no single inbox exceeds its cap. Send the drafted body, a short subject, your signature.
- 8. Wait.** Add a Wait node (e.g. 2 days) so replies have time to land before you check.
- 9. Get replies.** Add an IMAP Email (or Instantly reply) node to pull responses tied to each thread.
- 10. AI Agent - classify the reply.** Second LLM node. Paste the Classifier Prompt (Part 5). It returns JSON: {label, reason}. Label is one of interested / objection / not_now / unsubscribe.
- 11. Switch - route.** Add a Switch node keyed on label:
 - interested -> send yourself a Slack/email alert + your booking link
 - objection -> create a task for a human to reply
 - not_now -> add to a nurture list with a follow-up date
 - unsubscribe -> add to a suppression list, never contact again

Part 5 - The two prompts (copy-paste)

Draft Prompt (the outreach writer):

You write short B2B cold emails. Rules: under 90 words. One clear ask (a 15-minute call). No fluff, no "I hope this finds you well," no buzzwords. You MUST reference the one enrichment fact below so it reads as written-for-them, not mass mail. Plain, human, specific.

Lead: {{first_name}}, {{title}} at {{company}}.

Enrichment fact: {{signal}}.

Our offer: {{one_line_offer}}.

Output the email body only. No subject line, no sign-off block.

Classifier Prompt (the router's brain):

Classify this email reply into EXACTLY one label: interested, objection, not_now, unsubscribe.

Return only JSON: {"label": "...", "reason": "under 8 words"}

Rules: "interested" only if they want to talk or ask a real question. Vague politeness = not_now.

Any opt-out / remove-me language = unsubscribe. A pushback or "we already use X" = objection.

Reply: {{reply_text}}

Part 6 - Test before you unleash it

1. Run the workflow manually on 5 leads with Instantly set to send to YOUR OWN inboxes, not real leads.
2. Read all 5 drafts. Are they specific? Does the enrichment fact actually appear? Fix the prompt if not.
3. Reply to your own test emails as "interested", "not interested", "remove me" - confirm the classifier + Switch route each one correctly.
4. Only when all 5 look like something you'd send by hand, point it at real leads.

Part 7 - Go live + daily operation

- Turn the Schedule Trigger ON. It now runs every morning on its own.
- For the first 2 weeks, eyeball the "interested" bucket by hand each day and tune the classifier prompt from any misfires.
- Watch deliverability in Instantly. If open rates dip, slow your sending and let warmup keep running.
- Scale by adding mailboxes (each adds ~20-30 sends/day), never by over-sending existing ones.

Troubleshooting (the 5 things that break it)

- **Everything lands in spam.** You sent before warmup finished, or over-sent. Slow down, keep warmup on.
- **Emails read generic.** Your {{signal}} is weak or empty. Fix the enrichment step, not the prompt.
- **A broken {{company}} went out.** Your IF guardrail isn't catching placeholders. Tighten it.
- **Classifier mislabels replies.** Review by hand for 2 weeks, add the failing examples to the prompt.
- **It sends but nothing converts.** You built draft+send with no classify/route. That's a spam cannon, not an AI SDR. The routing is the whole point.

The catch, so you hear it from me: the model is just the engine. The moat is the workflow design, the enrichment quality, and the reply routing. A generic LLM dropped straight into sales scores **negative** on prospecting (GTM-Bench). The \$50 is the floor. The build quality above is what makes it actually work.

Want this mapped to your exact lead flow? DM me your current stack on LinkedIn and I'll show you the 2-3 nodes that'd save you the most time.

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